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MILITARY SPECIFICATION

INDICATORS AND TRANSMITTERS: LIQUID, TEMPERATURE AND PRESSURE

This specification has been approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers liquid quantity, pressure, and temperature indicators and transmitters for 24 volt (nominal) direct current (dc) electrical systems (see 6.1). The transmitters are of the electrical resistance type.

1.2 Classification. Indicators and transmitters shall be of the following units, types and classes, as specified (see 6.2):

Unit 1	- Indicators.
Type I	- Liquid quantity.
Type II	- Temperature.
Type III	- Pressure.
Unit 2	- Transmitters.
Type I	- Liquid quantity.
Class 1	- Float type.
Class 2	- Submerged tube type.
Type II	- Temperature.
Type III	- Pressure.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Materiel Readiness Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-380	- Plastic Molding Material, Methacrylate.
L-P-393	- Plastic Molding Material, Polycarbonate, Injection and Extrusion.
QQ-S-571	- Solder; Tin Alloy; Lead-Tin Alloy: and Lead Alloy.

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MIL-W-80	- Window, Observation, Acrylic Base, Antielectrostatic, Transparent (for Indicating Instrument).
MIL-E-13856	- Electrical Components for Automotive Vehicles; Waterproofness and Resistance to Temperature Cycling Tests, General Requirements for.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-P-81390	- Plastic, Molding Material, Polycarbonate, Glass Fiber Reinforced.

STANDARDS

FEDERAL

FED-STD-595	- Color.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-193	- Finishing Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	- Environmental Test Methods.
MS 24537	- Transmitter, Temperature Electrical Resistance 24 Volt DC.
MS 24538	- Transmitter, Pressure 24 Volt DC 60 PSI.
MS 24539	- Transmitter, Pressure 24 Volt DC 120 PSI.

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MS 24540	- Indicator, Pressure 24 Volt DC, Range 0 - 120 PSI.
MS 24541	- Indicator, Pressure 24 Volt DC, Range 0 - 60 PSI.
MS 24542	- Indicator, Temperature, Electrical Resistance 24 Volt DC, Range 160° - 320°F.
MS 24543	- Indicator, Temperature, Electrical Resistance 24 Volt DC, Range 120° - 240°F.
MS 24544	- Indicator, Liquid Quantity 24 Volt DC, Engine Fuel.
MS 500040(AT)	- Transmitter, Liquid Quantity, Float Type, Engine Fuel.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 First article. A first article sample of indicators and transmitters shall be produced prior to manufacture of the items in production quantities. Indicators and transmitters furnished under this specification shall be products represented by a first article sample that has been tested and has passed the first article inspection specified herein (see 4.3). Approval of the first article sample shall not relieve the contractor of his obligation to supply indicators and transmitters in production that are fully representative of those inspected as a first article sample. Any changes or deviations of production units from the first article sample shall be subject to the approval of the contracting officer.

3.2 Materials. Materials shall be as specified herein, in applicable specifications and on applicable MS standards or drawings. Materials not specifically designated, shall be suitable for use in indicators and transmitters operating over ranges specified without any change in physical or dimensional properties that would result in operation of the units falling outside of the limits specified herein (see 6.5).

3.2.1 Radioactivity. Indicators furnished under this specification shall not contain radioactive material.

3.2.2 Solder. Solder employed in making electrical connections shall

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conform to QQ-S-571. A rosin flux shall be used.

3.3 Construction. Indicators and transmitters shall be constructed to the form and dimensions specified on applicable MS standards or drawings, as specified (see 6.2).

3.3.1 Threaded surfaces. All threaded surfaces shall be in the inch system and shall conform to National Bureau of Standards Handbook H28.

3.3.2 Unit 1, indicators.

3.3.2.1 Dial.

3.3.2.1.1 Dial marking. The face of the dial shall be marked as shown on the applicable MS standard or drawing (see 6.2). Dial markings not covered by MIL-STD-130, such as numerals, letters and graduation marks shall have a minimum height of 1/8 inch. Numerals, letters and graduations shall be white conforming to color 37875 of FED-STD-595. Unless otherwise specified (see 6.2), numerals, graduations and other markings shall be clearly defined in white on a black background.

3.3.2.1.2 Fastening. The dial shall be secured to the body of the indicator in such a position that the marking will be legible when illuminated through the case windows from an outside source. The method of securing the dial shall be such that it will not become loose or turn when subjected to vibration or shock (see 4.5.10 and 4.5.11).

3.3.2.1.3 Pointer. The pointer shall be attached to the indicator movement and shall be damped to reduce oscillation from vibration. The pointer shall not touch the dial. The shape of the pointer shall not interfere with reading any point in the range of the dial. At least half of the visible portion of the pointer shall be white conforming to color 37875 of FED-STD-595.

3.3.2.1.4 Cover window. The cover window shall be treated on both sides with an electrostatic solution. The window shall be not less than 1/8 inch thick and shall be secured to the base in such a manner that moisture is excluded. The window shall have an Izod strength of not less than 0.3 foot-pounds. The window material shall retain transparency with age. The window material shall conform to MIL-W-80 or, if molded, it shall conform to type I, class 3 of L-P-380 or L-P-393.

3.3.2.2 Case. The case shall be constructed of steel suitable for the purpose, or plastic molding material conforming to L-P-393.

3.3.3 Unit 2, transmitters.

3.3.3.1 Mounting.

3.3.3.1.1 Type I. Type I transmitters shall be provided with a flange to

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permit attachment to fuel tanks. The diameters of the flange and boltholes, and bolthole spacing shall be as specified on the applicable MS standard or drawing.

3.3.3.1.2 Types II and III. Types II and III transmitters shall be provided with a male standard pipe thread connection of the size specified on the applicable MS standard or drawing.

3.4 Performance. Unless otherwise specified, indicators and transmitters covered by this specification shall meet performance requirements at all rated levels of temperature, pressure and electrical input, as applicable.

3.4.1 Calibration.

3.4.1.1 Unit 1, indicators. Unless otherwise specified on the applicable MS standard or drawing, the centerline of the pointer shall indicate the centerline of the appropriate dial graduation within plus or minus three angular degrees when tested in accordance with 4.5.2. When current is off, the pointer shall return to the extreme left position.

3.4.1.2 Unit 2, transmitters. When tested as specified in 4.5.3, the transmitter resistance measurements shall be in accordance with the applicable MS standard or drawing.

3.4.2 Voltage variation, indicators. When tested as specified in 4.5.4, indicators shall conform to 3.4.1.1 with an additional allowable tolerance of plus two angular degrees, minus four angular degrees.

3.4.3 Overload capacity, transmitters.

3.4.3.1 Type II. When tested as specified in 4.5.5.1, type II transmitters shall conform to 3.4.1.2 and shall evidence no mechanical or electrical damage, after exposure for one minute to a temperature of 350°F.

3.4.3.2 Type III. When tested as specified in 4.5.5.2, type III transmitters shall withstand a pressure of 200 percent of the maximum pressure specified on the applicable MS standard or drawing without evidence of mechanical or electrical damage. After this test, the transmitter shall conform to the requirements of 3.4.1.2.

3.4.4 Attitude error. When tested as specified in 4.5.6, indicators shall conform to 3.4.1.1 with an additional allowable tolerance of plus or minus two angular degrees.

3.4.5 Reverse connection capability, indicators. When tested as specified in 4.5.7, the indicators shall evidence no damage and shall conform to the requirements of 3.4.1.1 with an additional allowable tolerance of plus or minus one angular degree.

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3.4.6 Dielectric strength. When tested as specified in 4.5.8, the insulation of indicators and transmitters with metal cases only shall show no evidence of cracking or charring.

3.4.7 Endurance. After testing in accordance with 4.5.9, the indicators and transmitters shall meet the requirements of 3.4.1 with the following allowable tolerance variations:

a. Types I, II, and III indicators shall have an additional allowable tolerance of plus or minus two angular degrees.
b. Type I transmitters shall have an additional allowable tolerance of plus 2.0 ohms at empty and plus or minus 1.0 ohm at full.
c. Type II transmitters shall have an additional allowable tolerance of plus or minus one per cent of resistance specified at each calibration point.
d. Type III transmitters shall have an additional allowable tolerance of plus 0.5 ohm at zero indicator pressure and plus or minus 1.0 ohm at the other checkpoints specified on the appropriate drawing or MS standard.

3.4.8 Vibration resistance. Type I transmitters shall be tested in accordance with 4.5.10.1. Type II transmitters shall be tested in accordance with 4.5.10.2. Type III transmitters shall be tested in accordance with 4.5.10.3. Indicators shall be tested in accordance with 4.5.10.4. While indicators and transmitters are being vibrated, instruments shall be monitored to determine lapse of operation. Lapse of operation shall constitute test failure. Subsequent to the test, measurements shall be taken to determine whether the items meet the calibration requirements of 3.4.1 with added tolerance margins specified in 3.4.7. The instruments shall evidence no mechanical or electrical damage throughout the test.

3.4.9 Shock resistance. Types I, II and III indicators and types II and III transmitters shall be tested as specified in 4.5.11. After testing, indicators and transmitters shall evidence no mechanical or electrical damage and shall meet the calibration requirements of 3.4.1 with added tolerance margins as follows:

a. Types I, II and III indicators shall have an additional tolerance of plus or minus two angular degrees.
b. Type II transmitters shall have an additional tolerance of plus or minus one per cent of specified resistance at each calibration point.
c. Type III transmitters shall have an additional tolerance of plus 2.0 ohms at zero and plus or minus 1.0 ohm at the other checkpoints specified on applicable MS standard or drawing.

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3.4.10 Temperature influence.

3.4.10.1 Indicators. The indicators shall operate throughout an ambient air temperature range of minus 65 to plus 160F. When tested as specified in 4.5.12.1, indicators shall conform to 3.4.1 with an additional allowable tolerance of plus or minus five angular degrees.

3.4.10.2 Transmitters. The transmitters shall operate throughout an ambient air temperature of minus 65 to plus 160 F for type I, and minus 65 to plus 260 F for type III. When tested as specified in 4.5.12.2, transmitters shall conform to 3.4.1 with additional allowable tolerance of plus 1.0 ohm at empty or zero and plus or minus 1.0 ohm at the other checkpoints specified on the applicable MS standard or drawing.

3.4.11 Waterproofness.

3.4.11.1 Types I, II and III indicators and types II and III transmitters. When tested as specified in 4.5.13.1, types I, II and III indicators and types II and III transmitters shall meet the waterproofness requirements specified for type II, class 2 components in MIL-E-13856. When tested as specified in 4.5.13.2, types I, II and III indicators and types II and III transmitters shall show no evidence of leakage.

3.4.11.2 Type I, transmitter. When tested as specified in 4.5.13.3, type I transmitter components including floats, tubes and covers shall show no evidence of leakage.

3.4.12 Corrosion resistance. When tested as specified in 4.5.14, indicators and transmitters shall evidence no corrosion affecting operation and shall subsequently conform to 3.4.1.

3.4.13 Fungus resistance. When tested as specified in 4.5.15, indicators and transmitters shall evidence no fungus growth affecting operation and shall subsequently meet the requirements of 3.4.1.

3.5 Finish. The indicator bezel shall be lusterless camouflage forest green color per MIL-E-52798 or MIL-E-52835.

3.5.1 Preparation and painting. Cleaning, pretreatment, priming and painting shall conform to MIL-STD-193.

3.6 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.7 Workmanship. Workmanship shall be such as to assure a product

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free of rust, burrs, and scratches. All electrical connections shall be mechanically secured, before being soldered or brazed, to preclude the possibility of becoming loose due to vibration or shock. All solder cleaning agents and flux shall be noncorrosive. All traces of flux shall be chemically neutralized and removed after soldering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article sample (see 3.1), shall consist of four specimens of each type and class of unit to be tested. The specimens, properly marked with identifying information, shall be representative of the items proposed to be furnished to the Government. Unless otherwise specified, first article inspection shall be done by the contractor under Government surveillance and shall consist of examination for the defects specified in 4.4.2, and testing as specified in table I.

Table I. Order of first article testing.

Unit 1 - Indicators	
Specimen A:	
4.5.2	- Calibration.
4.5.4	- Voltage variation.
4.5.12.1	- Temperature influence.
4.5.9.1	- Endurance.
Specimen B:	
4.5.2	- Calibration.
4.5.6	- Attitude error.
4.5.11	- Shock.
4.5.13.1	- Waterproofness.

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Table I. Order of first article testing. Continue

Specimen C:	
4.5.8	- Dielectric strength.
4.5.2	- Calibration.
4.5.10.4	- Vibration.
4.5.14	- Corrosion resistance.
4.5.7	- Reverse connection capability.
Specimen D:	
4.5.2	- Calibration.
4.5.15	- Fungus resistance.
4.5.2	- Calibration.
Unit 2 - Transmitters	
Specimen A:	
4.5.3	- Calibration.
4.5.12.2	- Temperature influence.
4.5.9.2	- Endurance.
Specimen B:	
4.5.8	- Dielectric strength.
4.5.3	- Calibration.
4.5.10.1 (Type I)	- Vibration.
4.5.10.2 (Type II)	-
4.5.10.3 (Type III)	-
4.5.14	- Corrosion resistance.
Specimen C:	
4.5.3	- Calibration.
4.5.11	- Shock.
4.5.13.1 (Types II and III)	- Waterproofness
4.5.13.3 (Type I)	- Waterproofness
4.5.5	- Overload.
Specimen D:	
4.5.3	- Calibration.
4.5.15	- Fungus resistance.
4.5.3	- Calibration.

4.3.1 Failure. Failure of a first article sample to pass any of the tests specified herein may be cause for the Government to refuse to permit additional testing until the faults revealed by the testing have been corrected.

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4.4 Quality conformance inspection.

4.4.1 Sampling for units.

4.4.1.1 Lot formation. A lot shall consist of all indicators or transmitters of one type and class, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.1.2 Sampling for quality conformance examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.4.1.3 Sampling for quality conformance testing. From each lot that has passed the inspection specified in 4.4.2 a representative sample shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.4.2 Examination. Each indicator or transmitter selected in accordance with 4.4.1.2 shall be examined for conformance to applicable drawings and this specification. Examination shall be performed against the classification of defects and acceptable quality levels (AQL) specified in table II.

TABLE II. Classification of defects.

Categories	Characteristics	Method of inspection
Major:	AQL 1.0	
101 ------	Dimensions affecting interchangeability not within tolerance (see 3.3)	Gage
Minor:	AQL 2.5	
201 ------	Dimensions not affecting interchangeability not within tolerance (see 3.3)	Gage
202 ------	Defective painting (see 3.5)	Visual
203 ------	Improper marking (see 3.6)	Visual
204 ------	Faulty workmanship (see 3.7)	Visual

4.4.3 Classification of tests. Classification of tests shall be as follows:

a. Quality conformance tests (see 4.4.4).
b. Control tests (see 4.4.5).

4.4.4 Quality conformance tests. Samples selected in accordance with 4.4.1.3 shall be subjected to the tests specified in 4.5.3 or 4.5.4 and 4.5.13.2 or 4.5.13.3 as applicable using an AQL of 6.5 on the basis of per ent defective.

4.4.4.1 Failure. A failure rate exceeding 6.5 AQL shall be cause for

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rejection of the entire lot and may be cause for refusal of the Government to accept subsequent lots until it has been proven that the faults revealed by the test have been corrected. Individual failed samples shall not be resubmitted until defects have been corrected.

4.4.5 Control tests.

4.4.5.1 Sampling and procedure for control tests. Control test samples shall be selected at the rate of three of each 500 items produced of a particular unit, type, and class, except that not less than three nor more than six items shall be selected in any 30-day period. Samples shall be selected from a lot which has passed the examination specified in 4.4.2 and subjected to a separate series of tests as follows:

Item 1:	Calibration	4.5.2 or 4.5.3
	Vibration	4.5.10
Item 2:	Calibration	4.5.2 or 4.5.3
	Waterproofness	4.5.13.1 or 4.5.13.3
Item 3:	Calibration	4.5.2 or 4.5.3
	Temperature influence	4.5.12

4.4.5.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proven to the satisfaction of the Government that the faults revealed by the examination or test have been corrected.

4.5 Performance tests.

4.5.1 Conditions. Tests shall be conducted in a laboratory free of dust, drafts, fumes, and products of combustion. Room temperature shall be maintained at $77^{\circ} \pm 15^{\circ}$F. Test specimens shall be stabilized for one hour at this temperature before being subjected to tests. Readings shall be taken on indicators and from type II transmitters only after voltage has been applied for a minimum of five minutes. Indicators and type I and type III transmitters may be lightly tapped while readings are being taken.

4.5.2 Indicators (unit 1) calibration.

4.5.2.1 Apparatus. Apparatus shall include a test chamber with air temperature variable within a range from minus 65°F to plus 160°F, a protractor, a voltmeter, an ammeter, a decade resistance box, and a source of 28 ± 0.5 volt electrical energy.

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4.5.2.1.1 Calibration of test equipment. Test equipment resolution shall be such as to permit measurement of 10 percent of product tolerance. Instruments such as voltmeters, ammeters, and thermometers shall be of the laboratory type and shall be accurate to within ± 0.50 percent. The protractor shall be accurate to within ± 0.25 degrees and shall be free from error due to parallax. Records of calibration of test equipment, made at intervals sufficient to establish required accuracy, shall be made available to the Government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the Government.

4.5.2.2 Procedure. To determine conformance to 3.4.1.1, the indicator shall be mounted in a vertical position. The temperature shall be held at 77° ± 15°F. With the unit terminals connected to the voltage source and with resistance applied as specified on applicable MS standards or drawings, dial pointer readings shall be taken, by means of a precision protractor, at each calibration point.

4.5.3 Transmitter (unit 2) calibration.

4.5.3.1 Type I.

4.5.3.1.1 Apparatus. Apparatus shall include a fuel tank equipped with a fixture for attaching the unit, a wheatstone bridge or ohmmeter, and test liquid having a specific gravity of .721 ± 0.0036.

4.5.3.1.2 Procedure. To determine conformance to 3.4.1.2, the following procedure shall apply. Calibration shall be on a descending scale. With the tank filled with liquid to such a level that the float (class 1) or the tube (class 2) is at the full position, the transmitter shall be attached to the fixture installed in the tank. The resistance of the unit shall then be measured by means of a wheatstone bridge or ohmmeter. The test shall be repeated with the tank empty. Class 1 (float type) transmitters may be tested with the float at "full" and "empty" positions without being installed in the fuel tank.

4.5.3.2 Type II.

4.5.3.2.1 Apparatus. Apparatus shall include an oil tank provided with means for controlling the oil temperature and recirculating the oil, a laboratory type milliammeter, a laboratory type voltmeter, a supply of engine lubricating oil, and a source of 28 ± 0.5 volt dc electrical energy.

4.5.3.2.2 Procedure. To determine conformance to 3.4.1.2, the following procedure shall apply. Calibration for type II transmitters shall be on

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an ascending scale. The unit shall be immersed in a tank filled with oil, starting at the lowest temperature specified on the applicable MS standard or drawing. The oil shall then be circulated at a minimum rate of 10 feet per minute past the unit. The unit shall subsequently be subjected to each of the temperatures specified on the applicable MS standard or drawing and the internal resistance at each temperature determined.

4.5.3.3 Type III.

4.5.3.3.1 Apparatus. Apparatus shall include a dead weight tester or an oil or air tank with provisions for controlling the pressure, a precision pressure gage, and a wheatstone bridge or ohmmeter.

4.5.3.3.2 Procedure. To determine conformance to 3.4.1.2, measurements shall be recorded in accordance with following procedure. Calibration shall be on an ascending scale, starting at the lowest pressure. The unit shall be subjected to each of the values of pressure specified on the applicable MS standard or drawing for one minute and the internal resistance measured with the wheatstone bridge or ohmmeter.

4.5.4 Voltage variation. To determine conformance to 3.4.2, the indicator shall be subjected to the test specified in 4.5.2, except that the unit shall be connected first to a 20 volt dc and then to a 32 volt dc test circuit. Readings shall be taken at each value.

4.5.5 Transmitter (unit 2) overload capacity.

4.5.5.1 Type II. To determine conformance to 3.4.3.1, the transmitter shall be examined after the following test. The type II transmitter shall be tested as specified in 4.5.3.2 except that it shall be held in a 350°F oil bath for one minute prior to testing.

4.5.5.2 Type III. To determine conformance to 3.4.3.2, the type III transmitter shall be subjected to an oil or air pressure of 200 per cent of that pressure specified for full scale reading on the applicable indicator for one minute, then tested as specified in 4.5.3.3, and examined.

4.5.6 Attitude error. To determine conformance to 3.4.4, the indicator shall be mounted in such a manner that the dial face can be positioned as follows:

a. 45 degrees clockwise from normal (vertical plane).
b. 45 degrees counterclockwise from normal (vertical plane).
c. Face up (horizontal plane).
d. Face down (horizontal plane).

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With the exception of the mounting position, the procedure specified in 4.5.2.2 shall be followed. Dial pointer readings shall be taken for each of the four dial face positions.

4.5.7 Reverse connection capability, indicators. To determine conformance to 3.4.5, the power source and signal connections to indicators shall be reversed for a period of 2 hours. A 28 ± 0.5 volt dc source shall be connected to the signal terminal and the lowest resistance specified on applicable MS standard or drawings shall be applied to the power terminal on the indicators.

4.5.8 Dielectric strength for indicators and transmitters with metal cases only. To determine conformance to 3.4.6, insulation shall be examined after the following test. With the lead wires within the indicator or transmitter cases disconnected from ground or ground terminal, the ungrounded winding or circuit shall be subjected to a potential of 440 volts root mean square (rms) at a frequency of 60 cycles per second (cps) applied between the case and ungrounded terminal or winding for one minute. This test to be conducted prior to final assembly for sealed items.

4.5.9 Endurance.

4.5.9.1 Unit 1, indicators. To determine conformance to 3.4.7, indicators shall be subjected to the test specified in 4.5.2, after having been subjected to the following cycling period. While electrically connected to the applicable transmitter or an equivalent resistance, the indicator shall be subjected to full scale deflection and return for a total of 200,000 cycles. One cycle of operation shall consist of one complete oscillation from extreme left to extreme right and return to extreme left. Cycling shall be at such rate as to produce full scale deflection of the indicator pointer.

4.5.9.2 Unit 2, transmitters. To determine conformance to 3.4.7, transmitters shall be subjected to the test specified in 4.5.3, after having been subjected to the following cycling period. The transmitter shall be tested for endurance for a period of 100,000 cycles (10,000 cycles for type II transmitters) corresponding to full scale deflection and return as indicated on the applicable indicator. Cycling shall be at a rate which will produce full scale deflection on the associated indicator pointer. Transmitters may be tested concurrently with their applicable indicators.

4.5.10 Vibration.

4.5.10.1 Vibration, type I transmitter. To determine conformance to 3.4.8, type I transmitters shall be mounted to simulate actual mounting in

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use. The units shall then be vibrated in the direction of each of the three major axes as specified in method 201 of MIL-STD-202. At the end of each vibration period along each of the three major axes and while the units are still being vibrated, the units shall be subjected to the test specified in 4.5.3.1.

4.5.10.2 Vibration, type II transmitter. To determine conformance to 3.4.8, type II transmitters shall be mounted in a test fixture simulating actual mounting in use and shall be connected in a series circuit with a 28 ± 0.5 volt dc power source and applicable indicator. Care shall be taken to see that the mounting is free from resonance over the frequency range. While energized, the transmitter shall be subjected to simple harmonic motion having an amplitude of 0.03 inches or 50 gravity units (G) peak, whichever is less. Tolerance of plus or minus ten percent is permissible for the amplitude. The vibrational frequency shall be varied over the range from 10 to 3600 cps. Rate of frequency change shall be logarithmic. When there is no provision for logarithmic cycling, other automatic cycling rates of frequency change may be used. The vibrational cycle from 10 to 3600 and back to 10 cps shall be accomplished in 20 ± 2 minutes. This scanning cycle shall be repeated three times in order that critical frequencies (see 6.4) may be identified, recorded and checked. After the scanning cycles, the transmitter shall be vibrated at the critical frequency for two hours. If there is more than one critical frequency, the two-hour period shall be divided equally between the critical frequencies. If there are more than three critical frequencies, the three most critical shall be selected and the transmitter vibrated for 40 minutes at each of the frequencies selected. If no critical frequency is identified, the specimen shall be vibrated at 50G acceleration, with frequency cycled from 1400 to 3400 and back to 1400 cps. Rate of change of frequency shall be logarithmic or, where there is no provision for logarithmic cycling, other automatic cycling rates of frequency change may be used. Time for each cycle shall be 20 ± 2 minutes. Duration of the test shall be two hours (six complete cycles). At the end of the two hour vibrational period, the transmitter shall be subjected to the applicable test specified in 4.5.3. This test procedure shall be accomplished along each of three mutually perpendicular axes. Total vibrating time shall be nine hours (three hours along each axis).

4.5.10.3 Vibration, type III transmitter. To determine conformance to 3.4.8, the transmitter shall be tested in accordance with test condition A of method 204 of MIL-STD-202. During the test, the transmitter shall be rigidly mounted as in normal operating position and shall be connected in a series circuit with a 28 ± 0.5 volt dc power source and applicable indicator.

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4.5.10.4 Vibration, indicators. To determine conformance to 3.4.8, indicators shall be mounted vertically and vibrated in each of the three major axes as specified in method 201 of MIL-STD-202. At the end of each vibration period the indicators shall be subjected to the test specified in 4.5.2.

4.5.11 Shock. To determine conformance to 3.4.9, the units indicated shall be subjected to the applicable test specified in 4.5.2 or 4.5.3, after having been subjected to the following test. Types I, II, and III indicators and types II and III transmitters shall be mounted to simulate actual mounting in use. The units shall be subjected to the shock test as specified in method 213 of MIL-STD-202 in such manner than an impact of 100G will be obtained. The test shall be repeated 10 times and applied on the vertical axis while the test unit is in its normal mounting orientation. An optional method may be used as specified in 4.5.11.1 in such manner that an impact value of 50G will be obtained.

4.5.11.1 Shock, optional test procedure.

4.5.11.1.1 Apparatus. The shock-testing apparatus specified herein and shown on figures 1 through 8 inclusive, shall be used. The main structure consists of a cast-iron base to which upright channels are securely bolted. The tops of the upright channels are secured to a top channel which also supports rods to guide the movable carriage assembly. The other ends of the guide rods are screwed into the base. The carriage assembly carries on its underside a stiff calibrated spring resiliently mounted, but at the same time having helical compression springs sufficiently strong to hold the calibrated spring in contact with the effective cylindrical bearings. The cast-iron base shall be single piece constructed so that it has no practical yielding. The prime requisite in the construction of the movable carriage assembly is rigidity. For this reason 1/4 inch thick steel shall be used. After being securely fastened together with screws, the carriage parts shall be welded so as to give the equivalent of single-piece construction.

4.5.11.1.2 Mounting accessories. Mounting accessories, which will assure rigid fastening of specimens to the carriage assembly, shall be used. Since special tests may require special mounting fixtures, the carriage test plate shall be provided with a number of extra holes. The top of the carriage frame shall be provided with a holding bolt to facilitate the rigid clamping necessary.

4.5.11.1.3 Stationary anvil. The stationary anvil shall be made of hardened tool steel with a radius of curvature of 1 1/2 inches and shall be bolted tightly to the base.

4.5.11.1.4 Calibrated spring. The calibrated spring shall be made of

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first-quality carbon tool steel, spring-tempered, and having a hardness of 85 to 95, Rockwell B scale. It shall be annealed and ground flat stock of the following composition:

	Percent
Carbon	1.08 to 1.18
Chromium	0.40 to 0.50
Manganese	.20 to .30
Silicon	.20 to .30
Phosphorous and sulfur	.025 maximum

The spring shall be 1 3/4 inches wide and approximately 0.150-inch thick, and shall have an overall length of 7 1/2 inches. The approximate spring constant (K) shall be between 5,000 and 5,500 pounds-per-inch deflection.

4.5.11.1.5 Calibration. The actual calibration shall be made on the completely assembled machine. Static loads up to 500 pounds shall be applied to the top of the carriage frame, and the spring deflections measured by means of a dial gage or height gage. The gage shall be mounted on the top surface of the cast-iron base as one reference surface, and measurements shall be taken at the center of the carriage test plate. The

$$\text{Spring constant (K)} = \frac{\text{load in pounds}}{\text{inches deflection}}$$

For a spring constant of 5,000, the deflection for a 500-pound load would be 0.100 inch.

4.5.11.1.6 Calibration curves. Use of the shock-testing mechanism is facilitated by plotting calibration curves for various weights of specimens. Ordinarily, parallel curves for specimen weights of 0.5 pounds and 2.0 pounds will be adequate, and intermediate weights can be estimated. Figure 8 shows such curves arbitrarily calculated on the basis of a mean spring constant (K) of 5,250 and a total carriage-and-mounting-fixture weight of 9 3/4 pounds.

4.5.11.1.7 Computation of acceleration (G). The following formula for obtaining the required acceleration or shock, in terms of gravity units (G), is utilized for variations of component weight and spring constant (K):

$$G = \sqrt{\frac{2hK}{W}}$$

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(When G is specified, the necessary height may be determined from:

$$h = \frac{G^2 w}{2K}$$

Where:

G = nominal value of acceleration (gravity units).
h = height from which carriage is dropped (inches).
K = spring constant (pounds-per-inch deflection).
w = weight of carriage and its test load (pounds).

4.5.11.1.8 Procedure. The shock-testing apparatus shall be mounted on a sturdy laboratory table or equivalent base and leveled before use. The specimen shall be rigidly mounted on the carriage. For each blow, the carriage shall be raised to the height necessary for obtaining the required acceleration and then allowed to fall. Means shall be provided to prevent the carriage from striking the anvil a second time.

4.5.12 Temperature influence.

4.5.12.1 Unit 1, indicators. To determine conformance to 3.4.10.1, the indicator shall be subjected to the test specified in 4.5.2, except that the ambient air temperature shall be stabilized at minus 65° ± 5°F and subsequently at plus 160° ± 5°F. The indicator shall be soaked for four hours in the test chamber until its temperature is stabilized at that of the ambient air. Readings shall be recorded at each temperature.

4.5.12.2 Unit 2, transmitters. To determine conformance to 3.4.10.2, transmitters shall be subjected to the applicable test specified in 4.5.3, except that the ambient temperature and the transmitter temperature shall be stabilized, and the resistance of the unit measured, at each of the following temperatures.

Type I	-	Plus 160° ± 5°F and minus 65° ± 5°F.
Type III	-	Plus 260° ± 5°F and minus 65° ± 5°F.

4.5.13 Waterproofness.

4.5.13.1 Waterproofness, first article and control tests. To determine conformance to 3.4.11.1, types I, II and III indicators and types II and III transmitters shall be subjected to the waterproofness test for type II, class 2 components specified in MIL-E-13856.

4.5.13.2 Waterproofness, acceptance test. To determine conformance to 3.4.11.1, types I, II and III indicators and types II and III transmitters shall be subjected to an internal pressure of six pounds per square inch (psi) for 5 minutes, while totally immersed in tap water. Observation for air bubbles from the interior of the component shall be maintained during the 5

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minute period. Bubbles which are the result of entrapped air on the various exterior surfaces of the component shall not be considered a leak.

4.5.13.3 Waterproofness, type I transmitter. To determine conformance to 3.4.11.2, floats, tubes and covers on type I transmitters shall be tested for leakage by immersion in the salt water solution of MIL-E-13856 at a temperature of not less than 200°F. Covers will be tested for leakage under a tank pressure of six psi when the cover is mounted on the tank. Considerations relating to bubble formation of 4.5.13.2 will apply.

4.5.14 Corrosion resistance. To determine conformance to 3.4.12, indicators and transmitters shall be subjected to the salt fog test specified in method 509 of MIL-STD-810, except that the test duration shall be 200 hours. When the corrosion test follows the vibration test, the additional tolerance margins specified in 3.4.7 shall be applicable. Subsequent to testing, indicators and transmitters shall meet the requirements of 3.4.1.1 or 3.4.1.2, as applicable.

4.5.15 Fungus resistance. To determine conformance to 3.4.13, indicators and transmitters shall be subjected to the fungus test specified in method 508 of MIL-STD-810, except that the test period shall be a continuous 90 days. Subsequent to testing, indicators and transmitters shall meet the requirements of 3.4.1.1 or 3.4.1.2, as applicable.

4.6 Inspection of packaging.

4.6.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to requirements of Section 5. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable packaging data sheet as furnished at the time of invitation for bids; for the level of preservation and packaging, and level of packing specified (see 6.2).

6. NOTES

6.1 Intended use. An indicator (dash unit) and a transmitter (sending unit), when connected by a single conductor cable (not supplied with the device) and appropriately grounded, comprise a pressure, temperature, or liquid quantity gage. These devices are used in military motor vehicles and

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other internal combustion motor applications to inform the operator of the following: Type I, the level of fuel in the supply tank(s); type II, the engine or transmission coolant, or other liquid, temperature; type III, the oil pressure in the engine or transmission lubricating system or the air pressure in the airbrake system.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Unit, type and class of item required (see 1.2).
c. Applicable MS standards or drawings (see 2.1 and 3.3).
d. If other than white markings on black background are required (see 3.3.2.1.1).
e. Applicable packaging standard and level of preservation and packaging (see 5.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, tests and approval of the first article (see 3.1).

6.4 Definitions.

6.4.1 Critical frequency. Critical frequency is a resonant frequency developed in the transmitter such that amplitude exceeds that of the applied vibration. The greater the amplitude the more highly loaded the transmitter and the more critical the vibration environment.

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - YD

Review activity:
Army - ER
DLA - GS

User activities:
Army - ME
Navy - MC

Preparing activity:
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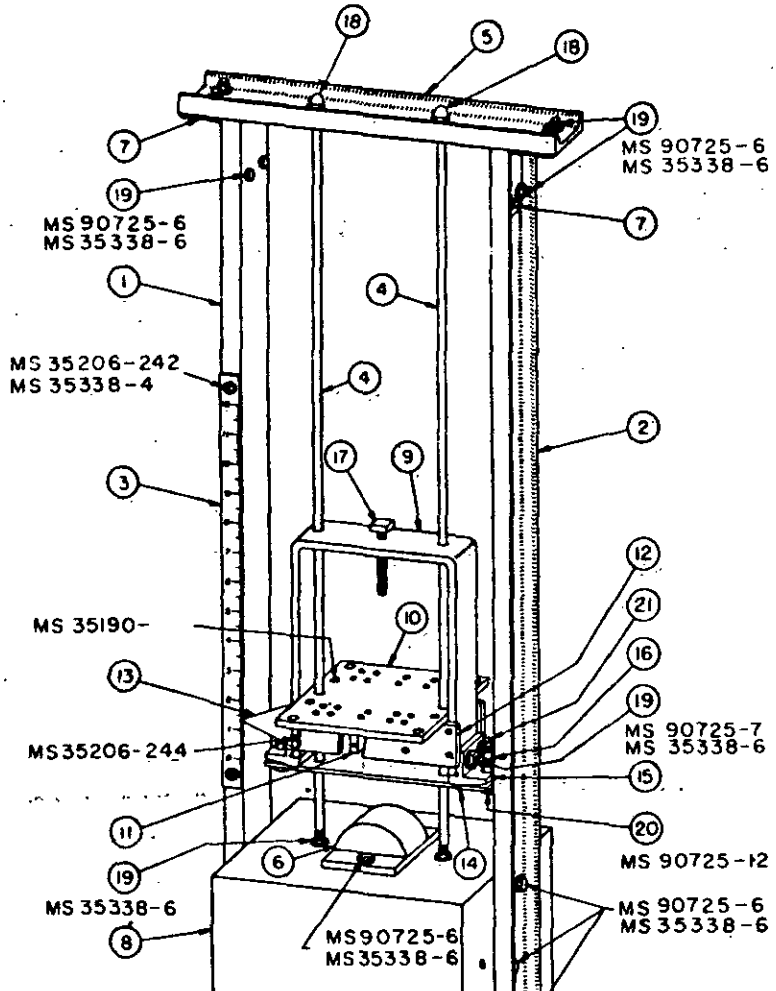


FIGURE 1. General assembly of shock-testing apparatus (see fig.7 for notes and bill of materials).

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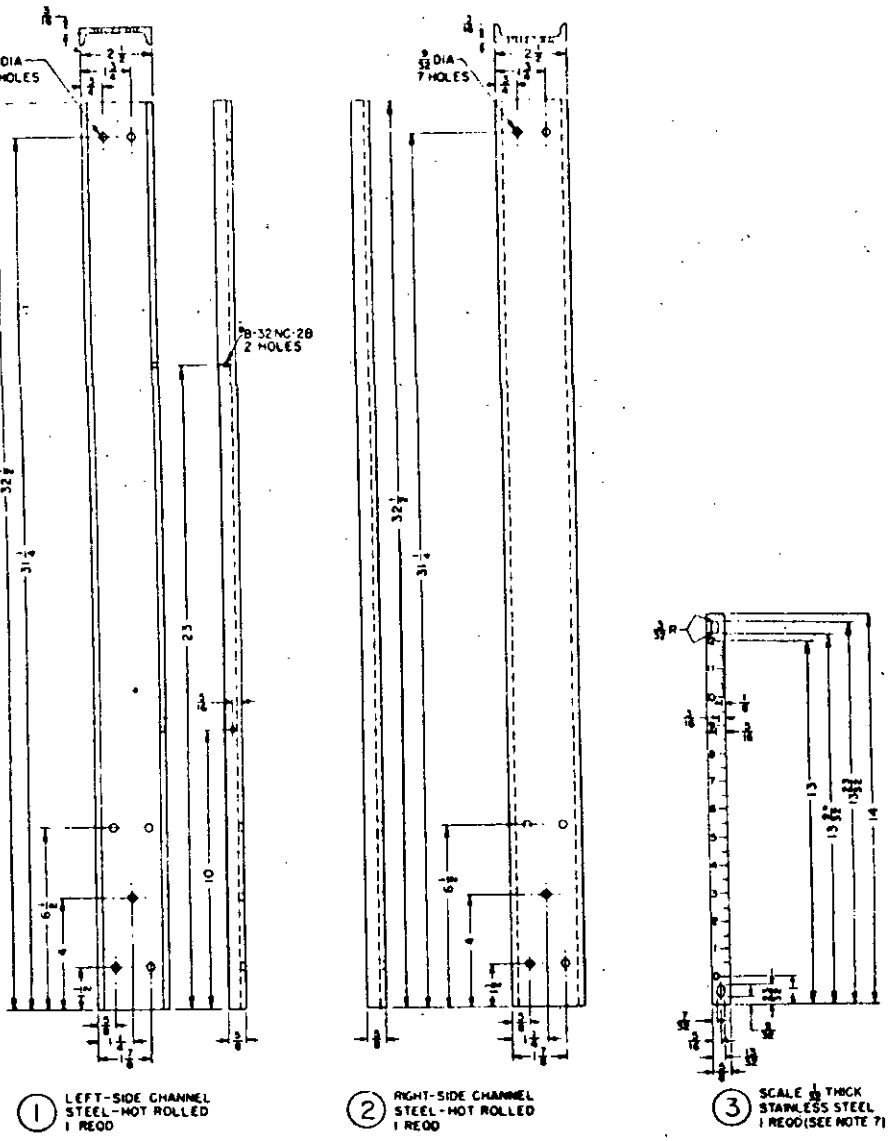


FIGURE 2. Main frame details(see fig. 7 for notes and bill of materials).

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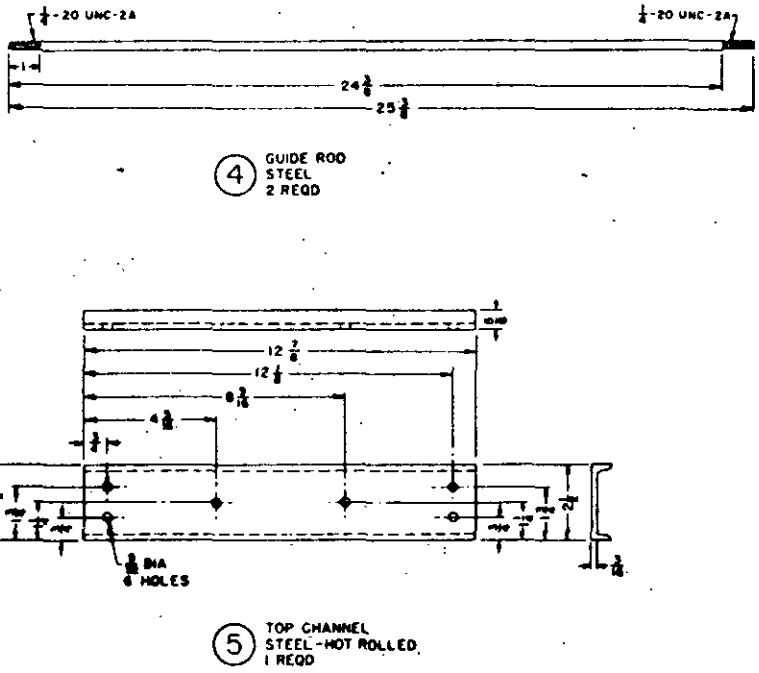


FIGURE 3. Main frame details (see fig. 7 for notes and bill of materials).

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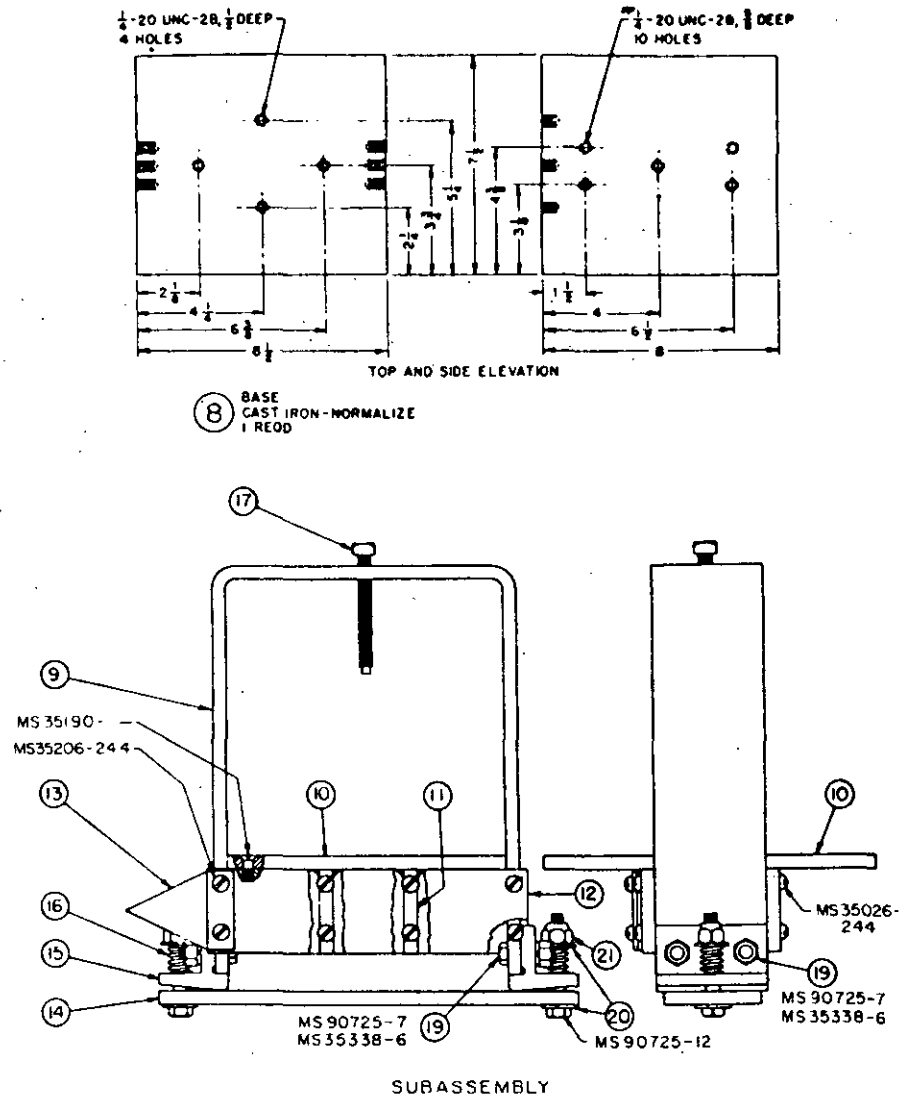


FIGURE 4. Carriage-assembly and base details (see fig. 7 for notes and bill of materials).

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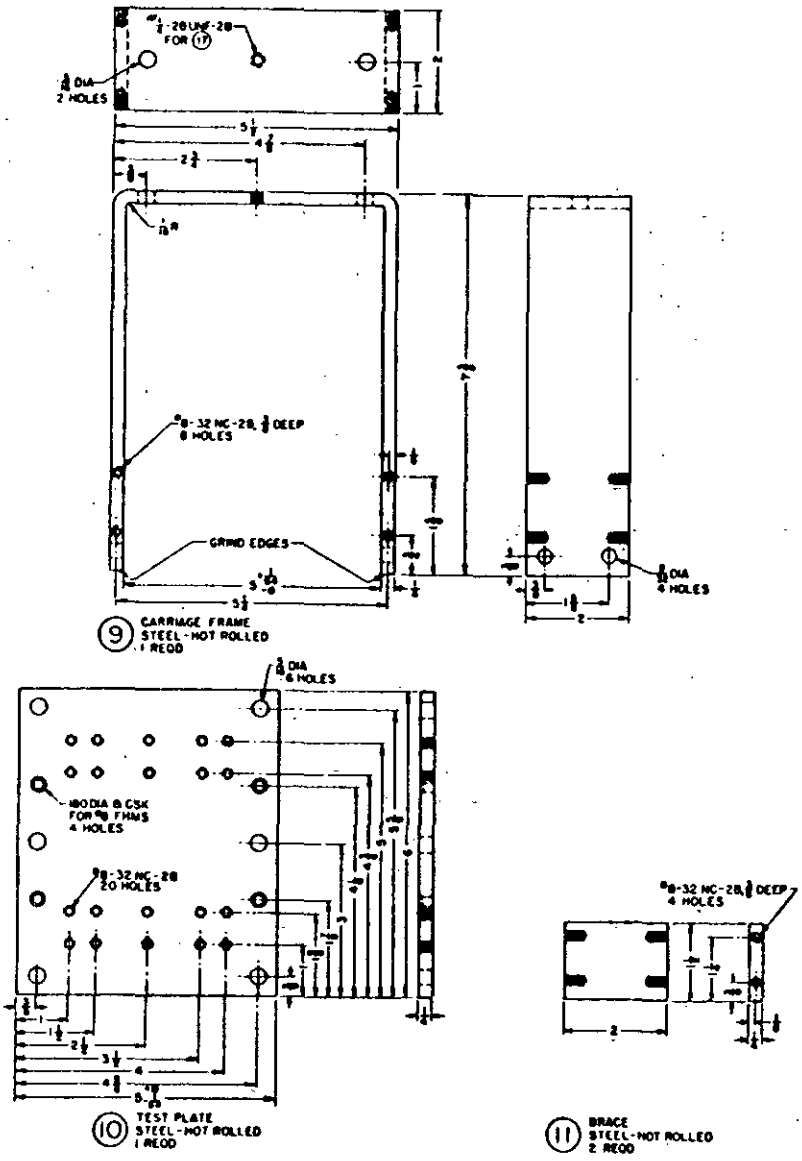


FIGURE 5. Carriage details (see fig. 7 for notes and bill of materials).

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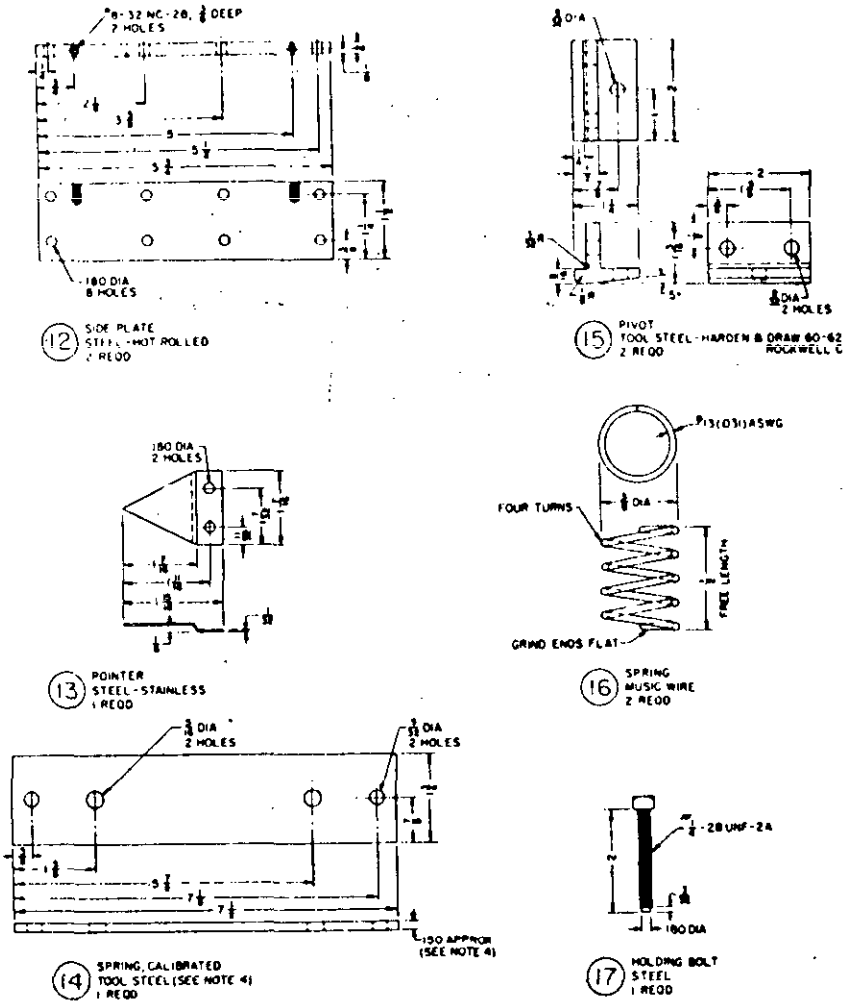


FIGURE 6. Carriage details (see fig. 7 for notes and bill of materials).

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Item No. or MS Part No.	Description	Quantity
1	Left-side channel	1
2	Right-side channel	1
3	Scale	1
4	Guide rod	2
5	Top channel	1
6	Anvil	1
7	Top angle	2
8	Base	1
9	Carriage frame	1
10	Test plate	1
11	Brace	2
12	Side plate	2
13	Pointer	1
14	Spring, calibrated	1
15	Pivot	2
16	Spring, compression	2
17	Squarehead bolt, No. ¼-28 UNF-2A x 2 long, steel, modified	1
18	Acorn nut, No. ¼-20 UNC-2B, steel	2
19	Hexagon nut, No. ¼-20 UNC-2B, steel	14
20	Flat washer, ¼, steel	4
21	Stop nut, per note 5	2
MS35206-242	Screw	2
MS35206-244	Screw	16
MS35190-51	Screw	4
MS90725-6	Bolt	20
MS90725-7	Bolt	4
MS90725-12	Bolt	2
MS35338-4	Lockwasher	2
MS35338-6	Lockwasher	26

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±1/64 on fractions, ±.005 on decimals and ±½° on angles.
3. Finish: The general assembly and the carriage shall be given a light spray coat of metal primer and then finished with a full spray coat of lusterless gray enamel. The mounting fixtures shall be cadmium or nickel plated. The spring, part 14, anvil, part 6, and pivots, part 15, shall be flash-dip nickeled.
4. Develop thickness experimentally so that spring constant will be 5,000 to 5,500 pounds-per-inch deflection when hardened and drawn.
5. Stop nut shall be catalog No. 29U040, as made by Elastic Stop Nut Corporation, Union, N.J., or equal.
6. The materials specified herein have been found satisfactory, but any suitable material may be used.
7. Scale shall be graduated in tenths of an inch. Fill black.

FIGURE 7. Notes and bill of materials.

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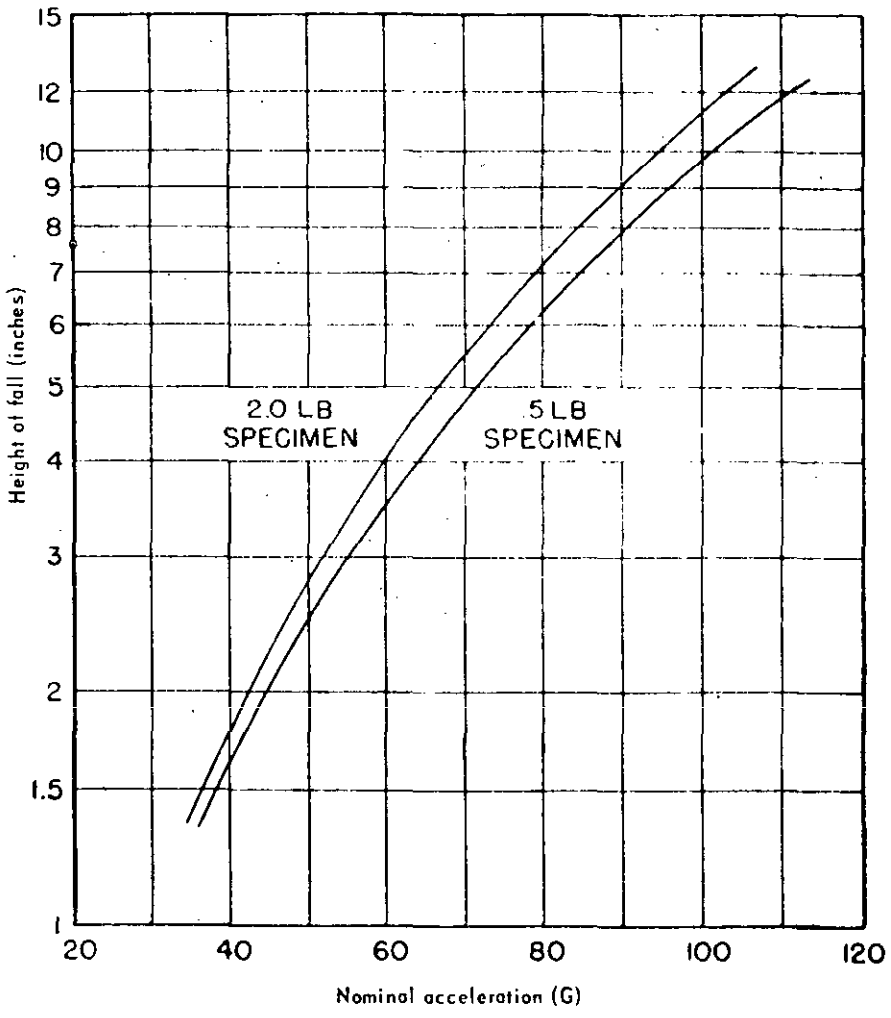


FIGURE 8. Typical calibration curves.

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